

Transport-Jaan — Module Status Index

Audit Date: 2026-06-15
Branch: ujith_new_dev
Stack: Laravel 10 + React 18 + Inertia.js + Tailwind CSS + PayHere

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Overall Readiness

Module	Readiness	UI	Backend API	Database
Vehicle Rental (Land · Air · Sea)	90 %	Done	Done	Done
Ticket Booking (Air · Train · Bus)	80 %	Done	Partial	Done
Courier Service (Domestic & Intl)	92 %	Done	Done	Done
Warehouse Booking	88 %	Done	Done	Done
Multi-Module (Bus · Train · Yacht)	55 %	Done	Partial	Done

Feature Areas per Module (Quick Reference)

Vehicle Rental — 8 Feature Areas

#	Feature Area	Status
1	Browsing & Discovery	Done Core done / Missing Advanced filters missing
2	Booking Flow	Done Core done / Missing Availability calendar missing
3	Post-Booking Management	Partial Cancellation done / Missing Modification missing
4	Cancellation Policies	Done Done
5	Reviews & Ratings	Done Done
6	Vendor — Vehicle Management	Done Core done / Partial Driver assignment partial
7	Pricing	Done Static / Missing Dynamic / commission missing
8	Insurance	Missing Not built

Ticket Booking — 5 Feature Areas

#	Feature Area	Status
1	Bus Ticket Booking	Done Core done / Missing Seat preference / QR missing
2	Train Ticket Booking	Done Core done / Partial Class selection partial
3	Flight / Air Ticket Booking	Missing Stub only — needs full build
4	Shared Booking Services	Done Validation / reference / PDF done
5	Operator / Vendor Portal	Partial Models exist / Missing No management UI

Courier Service — 13 Feature Areas

#	Feature Area	Status
1	Shipment Creation — Domestic	Done Done
2	Shipment Creation — International	Done Core done / Missing Customs docs missing
3	Favourite Recipients	Done Done
4	Payment	Done Core done / Partial Webhook signature unconfirmed
5	Client Shipment Dashboard	Done Core done / Missing Live map missing
6	Vendor Dashboard	Done Core done / Partial COD export UI unconfirmed
7	Shipping Labels	Done Core done / Missing QR scanning missing
8	Team Management	Done Core done / Missing Temp access UI missing
9	Security & Audit	Done Done
10	Email Notifications	Done Core done / Partial Suppression check unconfirmed
11	Pricing Governance	Done Service built / Partial Data seeding needed
12	Proof of Delivery & Field Ops	Missing Not built
13	Third-Party & API Integration	Partial Gateway partial / Missing Carriers not connected

Warehouse Booking — 7 Feature Areas

#	Feature Area	Status
1	Discovery & Browsing	Done Core done / Partial Some filters unconfirmed
2	Booking Flow	Done Core done / Partial Payment unconfirmed
3	Post-Booking Management	Done Cancellation done / Missing Modification missing
4	Reviews & Ratings	Done Done
5	Vendor — Unit Management	Done Core done / Missing Partial-capacity missing

6	Vendor Dashboard	Done Core done / Missing Payout split missing
7	Security & Advanced Features	Missing IoT / access control / insurance all missing

Multi-Module — 8 Feature Areas

#	Feature Area	Status
1	Journey Planning	Done Core done / Missing Connection validation missing
2	Vehicle / Service Search per Leg	Done Bus + Train / Partial Yacht unconfirmed
3	Selection & Cart	Done Cart done / Missing Seat selection / server total missing
4	Traveller Details	Done Core done / Missing Luggage missing
5	Payment	Partial Page exists / Missing PayHere call unconfirmed
6	Booking Confirmation	Done Pages done / Missing Unified PDF / vendor notify missing
7	Post-Booking Management	Partial Partial / Missing Leg cancel / modify missing
8	Transport-Type Integration	Partial Bus + Train partial / Partial Yacht unconfirmed

Recommended Launch Sequence

Phase 1 – Ready Now
└─ Vehicle Rental (Land · Air · Sea)
└─ Courier Service (Domestic & International)
└─ Warehouse Booking
Phase 2
└─ Ticket Booking (Bus + Train only, hide Flight tab)
└─ Multi-Module (Bus + Train legs only, remove Yacht until backend confirmed)
Phase 3
└─ Flight Ticket Booking (full build)
└─ Multi-Module Yacht leg

Each module document follows: Overview → Status Legend → Feature Breakdown (main feature → sub-features with Done / Partial / Missing status) → Critical Pre-Launch Fixes.

Module 1 — Vehicle Rental (Land · Air · Sea)

Launch Readiness: 90 %
Status: Near production-ready across all three vehicle types.

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Feature Breakdown

1. Browsing & Discovery

How clients find and explore vehicles across all three types (land, air, sea).

Sub-Feature	Status	Notes
Land vehicle listing page	Done	Paginated results with all land vehicles
Air vehicle listing page	Done	Separate listing for aircraft
Sea vehicle listing page	Done	Separate listing for boats/yachts
Search by brand / model	Done	Filter applied at listing level
Search by body type	Done	e.g. SUV, sedan, speedboat
Search by passenger capacity	Done	Min/max passenger count filter
Search by price range	Done	Price slider / min-max input
Search by transmission type	Missing	Not present in filter UI
Search by fuel type	Missing	Not present in filter UI
Search by mileage limit	Missing	Not present in filter UI
Vehicle detail page — land specs	Done	Full specification panel for land vehicles
Vehicle detail page — air specs	Done	Aircraft-specific specification panel
Vehicle detail page — sea specs	Done	Marine-specific specification panel
Media gallery (photos)	Done	Multiple photos per vehicle
Like / Favourite toggle	Done	Persisted per user in database
Star rating display	Done	Aggregated average shown on listing and detail

2. Booking Flow

The full journey from price quote to confirmed booking for all three vehicle types.

Sub-Feature	Status	Notes
Quote generation (land)	Done	Calculates cost from dates + duration + addons
Quote generation (air)	Done	Air-specific quote logic
Quote generation (sea)	Done	Sea-specific quote logic
Addon / extras selection	Done	Per-vehicle optional extras (GPS, child seat, crew, etc.)
Checkout page	Done	Quote summary + customer info form + payment entry
Customer info form	Done	Name, contact, special requirements
Booking type-specific fields	Done	Different fields shown for land vs air vs sea
Payment confirmation	Done	BookingPayment record stored on success
Booking summary / receipt page	Done	Reference, dates, and cost breakdown displayed
Availability check before booking	Missing	No date-overlap validation — double-booking is possible
Real-time availability calendar	Missing	No blocked-date widget on vehicle detail page

3. Post-Booking Management

What clients and vendors can do after a booking is confirmed.

Sub-Feature	Status	Notes
Booking history in client dashboard	Done	All bookings listed with status
Booking cancellation	Done	Cancellation policy evaluated before cancellation
Cancellation refund calculation	Done	Refund percentage based on cancellation window
Cancellation refund trigger (PayHere)	Partial	Policy evaluation exists; actual PayHere refund API call unconfirmed
Booking modification (date change)	Missing	Post-booking date-change flow does not exist
Driver assignment to booking	Missing	No UI for assigning a driver; vendor must handle off-system

4. Cancellation Policies

Rules that govern refund eligibility when a booking is cancelled.

Sub-Feature	Status	Notes
Configurable cancellation windows per vehicle	Done	Vendor sets cutoff periods

Configurable refund percentages per window	Done	e.g. 100 % if cancelled 7 days before, 50 % within 3 days
Policy displayed on checkout before payment	Done	Client sees policy before confirming
Policy enforcement on cancellation	Done	Applied when client initiates cancellation

5. Reviews & Ratings

Client feedback on vehicles after a completed rental.

Sub-Feature	Status	Notes
Star rating submission (1–5)	Done	Client rates after booking completion
Written comment submission	Done	Optional text review
Review listing on vehicle detail page	Done	All reviews shown below specs
Aggregated average rating	Done	Displayed on listing card and detail page

6. Vendor — Vehicle Management

Tools for rental vendors to manage their fleet.

Sub-Feature	Status	Notes
Add a new vehicle	Done	CRUD form for all three types
Edit vehicle details	Done	Update specs, pricing, description
Delete / remove a vehicle	Done	Remove from active listings
Set vehicle status (active / inactive / maintenance)	Done	Controls visibility to clients
Upload vehicle documents (registration, insurance)	Done	File storage per vehicle
Vehicle document PDF viewer (client side)	Partial	Upload works; inline PDF rendering on detail page unconfirmed
Upload vehicle photos	Done	Media gallery management
Maintenance log creation	Done	Log maintenance events per vehicle
Maintenance log tracking	Done	History of past maintenance records
Driver / crew profile management	Partial	Model and document exist; no UI to assign a driver to a booking
Driver self-registration	Missing	Drivers cannot register independently; vendor must add them manually

Vendor notification on new booking	Partial	Basic record saving exists; no email/push alert to vendor
View incoming bookings (vendor dashboard)	Partial	Dashboard exists; confirm all booking types appear correctly

7. Pricing

How vehicle rental prices are set and applied.

Sub-Feature	Status	Notes
Static price per vehicle	Done	Vendor sets a daily/hourly rate
Addon pricing	Done	Each extra has its own price
Dynamic / demand-based pricing	Missing	No surge or seasonal adjustment logic
Multi-currency support	Missing	Single currency only (LKR / PayHere default)
Vendor payout / commission split	Missing	No automated commission deduction or payout schedule

8. Insurance

Coverage options during the rental period.

Sub-Feature	Status	Notes
Insurance add-on during checkout	Missing	No third-party insurance quote or selection

Critical Pre-Launch Fixes

#	Fix	Impact
1	Availability blocking — Block booked dates so the same vehicle cannot be double-booked. Validate date overlap in <code>ClientBookingController::store()</code> .	High — data integrity
2	Cancellation refund trigger — Verify the PayHere refund API is actually called, not just the policy record updated.	High — financial
3	Vendor booking notification — Send email/notification to vendor when a new booking arrives so they can assign a driver.	Medium — operations
4	Vendor booking view — Confirm all three vehicle types (land, air, sea) appear in the vendor's incoming bookings dashboard.	Medium — operations

Module 2 — Ticket Booking (Air · Train · Bus)

Launch Readiness: 80 %

Status: Bus and Train are production-ready. Air/Flight booking is a stub only — no search, no seat map, no schedule system.

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Feature Breakdown

1. Bus Ticket Booking

End-to-end flow for searching and booking a bus seat.

Sub-Feature	Status	Notes
Search by origin station	Done	Station dropdown populated from DB
Search by destination station	Done	Station dropdown populated from DB
Search by travel date	Done	Date picker with past-date guard
Results listing	Done	Shows bus name, departure/arrival times, class, seats, price
Seat availability check	Done	Remaining seats validated before booking
Booking creation	Done	Stores passenger info, seat count, contact details
Seat number assignment	Partial	Availability counted but no individual seat number assigned to bus passenger
Window / aisle seat preference	Missing	No seat preference selection
Bus class selection (e.g. luxury / semi-luxury)	Missing	No class tier selection
Booking reference generation	Done	Cryptographic reference with embedded checksum
PDF ticket generation	Done	Downloadable ticket via TicketGenerationService
Email ticket delivery	Done	Ticket sent to passenger email on confirmation
Boarding pass / QR code	Missing	PDF only; no QR/barcode for digital scanning
Booking success page	Done	Shows reference, journey summary, and download link
Booking cancellation	Done	Policy-driven cancellation with refund eligibility check

Refund amount calculation	Done	Based on cancellation window
Actual PayHere refund reversal	Partial	Calculation done; PayHere API call unconfirmed
Booking modification (date / seat change)	Missing	Not supported post-booking

2. Train Ticket Booking

End-to-end flow for searching and booking a train seat.

Sub-Feature	Status	Notes
Search by origin station	Done	Station dropdown from DB
Search by destination station	Done	Station dropdown from DB
Search by travel date	Done	Date picker with past-date guard
Results listing	Done	Shows train, schedule, available seats, price
Individual seat selection	Done	Passenger picks a specific seat number from available pool
Seat conflict detection	Done	Two bookings cannot claim the same seat
Concurrent seat conflict (DB-level lock)	Partial	Application-level check exists; no DB unique constraint or pessimistic lock
Class selection (first / second / sleeper)	Partial	Seats are numbered but no class distinction shown in search or booking
Booking creation	Done	Stores passenger info and seat assignment
Booking reference generation	Done	Same cryptographic service as bus
PDF ticket generation	Done	Downloadable ticket
Email ticket delivery	Done	Ticket sent on confirmation
Boarding pass / QR code	Missing	PDF only
Booking success page	Done	Reference, journey, download link
Booking cancellation	Done	Policy evaluated before processing
Refund amount calculation	Done	Based on cancellation window
Actual PayHere refund reversal	Partial	Calculation done; PayHere API call unconfirmed
Booking modification	Missing	Not supported post-booking

3. Flight / Air Ticket Booking

Stub only — the booking flow does not exist yet.

Sub-Feature	Status	Notes
Flight schedule database	Missing	No flight schedule table or CRUD
Flight search by origin / destination / date	Missing	No search UI or backend query
Flight results listing	Missing	No results page
Seat map / seat selection	Missing	Not started
Airline / class selection	Missing	Not started
Booking creation	Partial	<code>FlightBookingController::store()</code> saves raw posted data only
Booking reference generation	Missing	Reference service not wired to flight flow
PDF ticket generation	Missing	Not wired
Email ticket delivery	Missing	Not wired
Baggage selection / allowance	Missing	Not started
Booking cancellation	Partial	Cancellation policy handler exists but no test path
Boarding pass / QR code	Missing	Not started
Airline partner / GDS integration	Missing	No Amadeus, Sabre, or equivalent API

4. Shared Booking Services

Infrastructure used across bus and train booking.

Sub-Feature	Status	Notes
Business validation (18 rules)	Done	Past dates, capacity, overbooking, seat conflict, duplicate bookings
Booking reference generator	Done	Collision-resistant, checksum-embedded
Ticket reference verification endpoint	Done	Validate a reference to confirm authenticity
Cancellation policy engine	Done	Calculates refund amount based on advance notice
PDF ticket generation service	Done	<code>TicketGenerationService</code> used by bus and train
Email rendering — Gmail / Outlook / mobile	Partial	Service exists; cross-client rendering not confirmed
Real-time seat availability push	Missing	Seat count checked at booking only; no live update for concurrent users
Baggage management	Missing	No allowance display or extra baggage fee selection

Multi-currency support	Missing	Single currency only
Foreign passenger passport field	Missing	No passport number field

5. Operator / Vendor Portal

Tools for bus and train operators to manage their services.

Sub-Feature	Status	Notes
Schedule management UI	Partial	Models and migrations exist; no operator UI to add / edit schedules
Station management UI	Partial	Models exist; no UI
Bus / train fleet management UI	Partial	Models exist; no UI
View bookings for own routes	Missing	Operators cannot log in to see their bookings
Capacity / manifest management	Missing	Not started

Critical Pre-Launch Fixes

#	Fix	Impact
1	Flight tab — Remove the flight tab from the UI at launch, or complete the minimum flight booking flow. A broken tab is worse than no tab.	High — UX
2	Refund reversal — Confirm CancellationPolicyService triggers the actual PayHere refund API, not only a DB status update.	High — financial
3	Concurrent seat lock — Add a DB-level unique constraint or pessimistic lock on seat assignment to prevent race conditions.	High — data integrity
4	Schedule seeding — Add at minimum a superadmin UI to create schedules; operators cannot use the system without schedules in the DB.	High — operations
5	Email template QA — Test ticket email rendering across Gmail, Outlook, and mobile before launch.	Medium — quality

Module 3 — Courier Service (Domestic & International)

Launch Readiness: 92 %

Status: Most complete module. Domestic and international flows, COD settlement, label generation, team management, and security auditing are all substantially implemented.

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Feature Breakdown

1. Shipment Creation — Domestic

Multi-step flow for sending a parcel within the country.

Sub-Feature	Status	Notes
Sender information form	Done	Name, address, contact fields
Recipient information form	Done	Name, address, contact fields
City lookup (province → district → city chain)	Done	Cascading dropdowns from normalized location DB
Postal code lookup	Done	Auto-fills based on city selection
Package weight input	Done	Captured for pricing
Package dimensions input	Done	Length × width × height
Declared value input	Done	For insurance / customs purposes
Service type selection (standard / express / same-day)	Done	Options shown based on availability
Shipment review / summary step	Done	Preview all details before payment
Bill / invoice PDF generation	Done	Downloadable from dashboard after booking
Favourite recipient pre-fill	Done	Saved recipients auto-fill the recipient form

2. Shipment Creation — International

Multi-step flow for sending a parcel to another country.

Sub-Feature	Status	Notes
International recipient form	Done	Country + address fields
Country lookup	Done	Normalized country list
Separate international pricing tier	Done	Different rate applied for international routes

Service type selection	Done	Standard / express options
Customs declaration form	Missing	No HS code lookup or customs value declaration
Commercial invoice generation	Missing	Not implemented

3. Favourite Recipients

Save and reuse recipient details for repeat shipments.

Sub-Feature	Status	Notes
Save new recipient as favourite	Done	Stored per user account
Pre-fill recipient form from saved list	Done	One-click fill on the shipment form
Edit saved recipient	Done	Update name, address, contact
Delete saved recipient	Done	Remove from favourites list

4. Payment

Payment flow for courier shipments.

Sub-Feature	Status	Notes
PayHere payment initiation	Done	Redirects to PayHere hosted page
Payment notification webhook (server-side)	Done	Handler exists in <code>CourierPaymentController</code>
PayHere MD5 signature validation	Partial	Webhook handler exists; signature verification must be confirmed
Return URL handling (success)	Done	Redirects client to success page
Cancel URL handling	Done	Handles PayHere payment abandonment
Retry failed payment	Done	Retry flow available from dashboard
Payment status tracking per shipment	Done	Status stored and displayed in dashboard
Duplicate payment prevention	Partial	No confirmed guard against rapid double-submit

5. Client Dashboard

Where clients monitor and manage their shipments, payments, and account.

Sub-Feature	Status	Notes
Shipments		
Shipment list with status badges	Done	All shipments shown with current status

Shipment detail view	Done	Full detail per shipment
Tracking event history per shipment	Done	Timeline of events on shipment detail
Live map tracking	Missing	Events stored but no map UI (Google Maps / Leaflet)
Bulk shipment upload (CSV/Excel)	Missing	No import tool
Documents		
Bill / invoice download	Done	Download PDF from shipment detail
Shipping label download	Done	Label available after booking
Wallet		
Wallet balance display	Missing	No wallet UI for client
Top up wallet	Missing	No top-up flow
Pay for shipment using wallet balance	Missing	Payment method not available
Wallet transaction history	Missing	No transaction list
Refund to wallet on cancellation	Missing	Refunds go back to original payment method only
Wallet statement download (PDF)	Missing	Not implemented
Favourite Recipients		
Saved recipient list	Done	View all saved recipients
Edit saved recipient	Done	Update name, address, contact
Delete saved recipient	Done	Remove from list
Notifications		
Shipment status change notification	Done	Shown on status updates
Email notification preference	Partial	Preference model exists; UI screen unconfirmed

6. Vendor Dashboard

Tools for courier operators to manage operations.

Sub-Feature	Status	Notes
Total shipments metric	Done	Count on dashboard home
Revenue metric	Done	Revenue total displayed
Pending / in-transit / delivered counts	Done	Status breakdown cards

Shipment list with search and filter	Done	Searchable, filterable table
Calendar view of shipment activity	Done	Activity calendar per vendor
Revenue by period chart	Done	Chart.js graphs for analytics
Payout summaries	Done	Financial summary view
COD settlement — batch creation	Done	Create a COD settlement batch
COD settlement — line items	Done	Per-shipment COD lines
COD settlement — reconciliation	Done	CourierCodSettlementReconciliationService (27 K)
COD settlement — export	Partial	Service class exists; UI trigger button unconfirmed
COD integrity monitoring alerts	Partial	Detection service exists; no dashboard widget for flagged incidents
Fraud detection alerts	Partial	CourierCodIntegrityMonitoringService runs checks; no real-time alert UI

7. Shipping Labels

Generation and management of parcel labels.

Sub-Feature	Status	Notes
Label creation from shipment data	Done	Auto-generated on shipment confirmation
Label template management	Done	Create, select, and apply label templates
Label size configuration (standard / large / thermal)	Done	Multiple size options
Print-ready label output	Done	Print-formatted output
Barcode / QR scanning for delivery agents	Missing	Label has reference text; no integrated scanning workflow

8. Admin Dashboard

Superadmin-level oversight of all courier operations across all vendors and clients.

Sub-Feature	Status	Notes
Overview Metrics		

Total shipments across all vendors	Done	Aggregated count in superadmin reports
Total revenue across all vendors	Done	Revenue totals in reports controller
Active vendors count	Done	Shown in admin panel
Active clients count	Done	Shown in admin panel
Pending / flagged shipments count	Partial	Basic counts exist; dedicated alert panel missing
Shipment Management		
View all shipments (all vendors)	Done	SuperAdmin reports controller covers courier shipments
Filter shipments by vendor	Done	Filter available in reports
Filter shipments by status	Done	Status filter available
Filter shipments by date range	Done	Date range filter available
Export shipment report (CSV/PDF)	Partial	Reports controller exists; export format confirmation needed
Vendor Management		
Approve new courier vendor	Done	Vendor approval workflow in place
Suspend / deactivate vendor	Done	Status update available via admin panel
View vendor performance metrics	Done	Per-vendor stats in reports
Commission rate management per vendor	Partial	Commission model exists; UI for setting per-vendor rates unconfirmed
COD & Financial		
View all COD settlement batches	Done	Superadmin COD operations (Phase 3 doc exists)
Approve COD settlement payouts	Done	Approval workflow in place
COD compliance export	Partial	Service exists; admin-facing trigger UI unconfirmed
Commission deduction overview	Partial	Calculation logic exists; dashboard widget unconfirmed
Wallet (Admin)		
View all client wallet balances	Missing	No wallet system built
View all wallet transactions	Missing	No wallet system built

Manually adjust wallet balance	Missing	Not implemented
Flag suspicious wallet activity	Missing	Not implemented
Pricing Governance		
Import / update pricing rules	Done	CourierPricingImportService (47 K)
View active pricing matrix	Partial	Service manages pricing; read-only UI unconfirmed
Security & Audit		
View security audit log (all vendors)	Done	CourierTeamSecurityAuditService logs available
View COD integrity incidents	Done	CourierVendorCodIntegrityIncident records visible
Manage email suppression list	Partial	Model exists; admin UI screen missing
Notifications		
Send broadcast notification to all clients	Missing	No broadcast tool
Send notification to specific vendor	Partial	Manual notification possible via dashboard

9. Wallet

Client-held balance used to pay for shipments without per-transaction payment processing.

Sub-Feature	Status	Notes
Balance Management		
Wallet creation on client registration	Missing	No wallet entity or table
Wallet balance display (client)	Missing	Not built
Wallet balance display (admin)	Missing	Not built
Top Up		
Top up via PayHere	Missing	No top-up flow
Top up via bank transfer	Missing	Not implemented
Minimum top-up amount validation	Missing	Not implemented
Top-up confirmation email	Missing	Not implemented
Payments		
Pay for shipment using wallet balance	Missing	Wallet not a selectable payment method
Insufficient balance error / fallback	Missing	No balance check at checkout

Split payment (wallet + card)	Missing	Not implemented
Refunds		
Refund to wallet on shipment cancellation	Missing	Refunds currently return to original payment method
Refund amount credited instantly	Missing	Not implemented
Transaction History		
Transaction list (top-ups, payments, refunds)	Missing	No transaction table or UI
Filter transactions by type / date	Missing	Not implemented
Wallet statement download (PDF)	Missing	Not implemented
Admin Controls		
View any client wallet balance	Missing	Not implemented
Manually credit / debit a wallet	Missing	Not implemented
Freeze a wallet	Missing	Not implemented
Wallet transaction audit log	Missing	Not implemented

10. Team Management

Multi-user access for vendors with role-based permissions.

Sub-Feature	Status	Notes
Add team member	Done	Vendor invites staff
Remove team member	Done	Revoke access
Role-based permission assignment	Done	Different access levels per member
Temporary access grant	Partial	Model and service built; no UI to issue a grant — programmatic only
Temporary access expiry	Done	Grants expire automatically
Sensitive-action approval workflow	Done	High-impact actions require second approval
Trusted-device registration	Done	Devices flagged as trusted per user
Temporary access grant UI	Missing	No screen for manually issuing temporary access

11. Security & Audit

Protection and logging of sensitive operations.

Sub-Feature	Status	Notes
Security audit log (all sensitive actions)	Done	Actor, timestamp, IP logged via <code>CourierTeamSecurityAuditService</code>
Access review certifications	Done	Periodic re-confirmation of access rights
COD integrity incident log	Done	<code>CourierVendorCodIntegrityIncident</code> model records anomalies
Session security service	Done	<code>CourierSessionSecurityService</code> manages session state

12. Email Notifications

Automated emails sent to clients and vendors.

Sub-Feature	Status	Notes
Shipment creation confirmation email	Done	Sent to client on creation
Status change notification email	Done	Sent on each status update
Email dispatch lifecycle management	Done	<code>CourierCustomerEmailDispatchService</code> (31 K)
Email delivery status webhook	Done	<code>CourierEmailDeliveryWebhookController</code> handles provider callbacks
Email suppression list (opt-out)	Partial	Model and logic exist; no admin/vendor UI to view or manage the list
Suppression check before every send	Partial	Must confirm service always checks suppression before dispatch

13. Pricing Governance

How courier rates are managed across zones and service types.

Sub-Feature	Status	Notes
Pricing import / management	Done	<code>CourierPricingImportService</code> (47 K) supports zone + weight-band + service-type rules
Pricing data seeded in DB	Partial	Service exists; confirm pricing rules have been loaded before first shipment
Vendor rate management UI	Partial	Import service is backend; UI completeness unconfirmed

14. Proof of Delivery & Field Operations

Last-mile delivery capture.

Sub-Feature	Status	Notes
Proof of delivery photo upload	Missing	Not implemented
Recipient signature capture	Missing	Not implemented
Driver / delivery agent portal	Missing	No mobile-optimized field interface

15. Third-Party & API Integration

External carrier and partner connections.

Sub-Feature	Status	Notes
API gateway controller	Partial	CourierServiceApiGatewayController (146 lines) exists; API key issuance UI and documentation unconfirmed
Third-party carrier integration (DHL, FedEx, etc.)	Missing	No external carrier API connected
Insurance add-on	Missing	No shipment insurance option during checkout

Critical Pre-Launch Fixes

#	Fix	Impact
1	PayHere webhook signature — Confirm MD5 hash validation in webhook handler to prevent spoofed payment notifications.	High — financial security
2	Duplicate shipment guard — Block rapid double-submits on the create-shipment form from creating two shipments and two charges.	High — financial
3	Pricing data seeding — Run the CourierPricingImportService import with real pricing data before the first shipment can be priced.	High — operations
4	Email suppression check — Confirm CourierCustomerEmailDispatchService always queries CourierEmailSuppression before sending.	Medium — compliance
5	COD cycle test — Run end-to-end COD batch creation → settlement → payout export before enabling COD for any vendor.	Medium — financial

Module 4 — Warehouse Booking

Launch Readiness: 88 %

Status: Core booking flow, vendor management, and client dashboard are complete. Payment integration and a few integration gaps remain.

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Feature Breakdown

1. Discovery & Browsing

How clients find and explore warehouse units.

Sub-Feature	Status	Notes
Warehouse listing page	Done	All available units displayed
Filter by location	Done	City / region filter
Filter by storage category (dry, cold, hazmat)	Done	Category filter in listing
Filter by unit size	Done	Size range filter
Filter by price range	Done	Min/max price filter
Filter by climate-controlled flag	Partial	UI component exists; backend query filter unconfirmed
Filter by hazmat certification	Partial	UI component exists; backend query filter unconfirmed
Unit detail page — photos	Done	Photo gallery per unit
Unit detail page — amenities list	Done	CCTV, forklift access, loading dock, etc.
Unit detail page — specification panel	Done	Dimensions, capacity, storage type
Unit detail page — documents	Done	Fire safety certs, operating licences
Like / Favourite toggle	Done	Persisted per user in database
Star rating display	Done	Aggregated average on listing and detail

2. Booking Flow

End-to-end process for reserving a warehouse unit.

Sub-Feature	Status	Notes
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Category selection step	Done	Client selects storage type before listing
Unit listing after category selection	Done	Filtered listing shown
Unit detail view before booking	Done	Full detail page accessible
Date and time range selection	Done	Start date/time + end date/time picker
Hourly pricing calculation	Done	Cost calculated per hour
Daily pricing calculation	Done	Cost calculated per day
Monthly pricing calculation	Done	Cost calculated per month
Availability check before booking	Done	Checks existing bookings before allowing creation
Concurrent availability race condition guard	Partial	Application-level check; no DB-level lock on date-range overlap
Customer info form	Done	Name, contact, requirements
Booking summary / receipt page	Done	Full cost breakdown displayed
Payment (digital)	Partial	Payment status field exists in DB; PayHere integration for warehouse unconfirmed
Payment (manual / cash)	Partial	If cash is intended, UI must communicate this clearly

3. Post-Booking Management

Actions available to clients after a booking is confirmed.

Sub-Feature	Status	Notes
Booking history in client dashboard	Done	All bookings listed
Booking cancellation	Done	Cancellation policy evaluated before processing
Cancellation refund calculation	Done	Based on configured cancellation window
Cancellation refund trigger	Partial	Calculation done; actual refund action unconfirmed
Booking extension (add more time)	Missing	Not supported
Booking shortening	Missing	Not supported
Multi-unit booking in one checkout	Missing	One unit per booking flow only

4. Reviews & Ratings

Client feedback after a completed storage booking.

Sub-Feature	Status	Notes
Star rating submission (1–5)	Done	Available after booking completion
Written comment submission	Done	Optional text review
Review listing on unit detail page	Done	All reviews shown
Aggregated average rating	Done	Displayed on listing card and detail page

5. Vendor — Unit Management

Tools for warehouse vendors to manage their storage units.

Sub-Feature	Status	Notes
Add a new warehouse unit	Done	CRUD form
Edit unit details	Done	Update specs, pricing, description
Delete / remove a unit	Done	Remove from listings
Upload unit photos	Done	Multiple images per unit
Define amenities	Done	Select from amenity list
Attach compliance documents	Done	File upload per unit
Set unit status (active / inactive / under review)	Done	Controls client visibility
Unit approval workflow	Done	New units sent to superadmin for approval before going live
Rejected unit cannot be booked	Partial	Workflow exists; confirm rejected units are blocked from booking
Set pricing (hourly / daily / monthly)	Done	Pricing fields per tier
Partial-capacity booking (e.g. 200 of 1 000 sq ft)	Missing	Entire unit is booked as a whole; no partial model

6. Vendor Dashboard

Analytics and management tools for warehouse vendors.

Sub-Feature	Status	Notes
Booking list with search and filter	Done	Searchable, filterable table
Reservation confirmation (from dashboard)	Done	Vendor can confirm a pending reservation

Reservation cancellation (from dashboard)	Done	Vendor can cancel a reservation
Total bookings metric	Done	Shown on dashboard home
Revenue by period chart	Done	Chart.js graphs
Occupancy rate chart	Done	Visual occupancy analytics
Calendar view of bookings	Done	Visual schedule per unit
iCal / Google Calendar export	Partial	WarehouseCalendarController exists; bidirectional sync unconfirmed
Client management	Done	View client details and booking history
Send notification to client	Done	Manual notification dispatch
Auto-notification on new booking	Partial	Controller exists; automatic email/SMS on new booking unconfirmed
Financial tracking — payment records	Done	Payment status and amounts logged
Financial tracking — expense logging	Done	Expense records per unit
Vendor payout / commission split	Missing	No automated commission deduction or payout

7. Security & Advanced Features

Features related to physical access and asset protection.

Sub-Feature	Status	Notes
Climate / temperature monitoring	Missing	No IoT data feeds for cold-storage units
Humidity / sensor alerts	Missing	Not implemented
Key-card / PIN access issuance	Missing	No access-control integration
Entry log for clients	Missing	No who-entered-when display
Insurance for stored goods	Missing	No insurance option during checkout
Inventory / SKU tracking	Missing	No inventory management for goods stored inside
Automated recurring invoice (monthly contracts)	Missing	Invoicing is manual

Critical Pre-Launch Fixes

#	Fix	Impact
1	Payment step — Clarify and implement payment. If digital, wire PayHere; if manual/cash, add clear instructions in the UI.	High — financial
2	Concurrent booking race condition — Add a DB-level lock or unique constraint on unit + date-range overlap.	High — data integrity
3	Approval workflow test — Run the full cycle: vendor submits → superadmin approves/rejects → confirm rejected units cannot be booked.	High — operations
4	Cancellation refund trigger — Verify policy logic triggers an actual refund action, not only a DB status update.	High — financial
5	Vendor notification — Send email to vendor when a client books their unit.	Medium — operations

Module 5 — Multi-Module Booking (Bus · Train · Yacht)

Launch Readiness: 55 %

Status: Journey-planning skeleton and UI are complete. Backend integration with individual transport providers is partial. Several critical security and operational gaps remain.

Status Legend

Symbol	Meaning
Done	Fully implemented and working
Partial	Partially implemented — logic or UI incomplete
Missing	Not implemented — needs to be built

Feature Breakdown

1. Journey Planning

Building a multi-leg trip from scratch.

Sub-Feature	Status	Notes
Multi-leg journey entry form	Done	Add origin, destination, date, and transport mode per leg
Add unlimited legs to a journey	Done	No hard cap on number of legs
Journey record saved to DB	Done	Stored in <code>MultiModelJourney</code> with linked <code>MultiModelLeg</code> rows
Journey retrieved across steps	Done	State carried through all subsequent pages

Connection time validation between legs	Missing	No check that arrival of leg N allows connection to departure of leg N+1
Itinerary optimization suggestions	Missing	No engine to recommend the best combination of options
Auth gate at journey-plan step	Partial	Unclear if unauthenticated users are blocked here or only at checkout

2. Vehicle / Service Search per Leg

Finding available transport for each individual leg.

Sub-Feature	Status	Notes
Bus schedule search per leg	Done	Available buses loaded by origin/destination/date
Train schedule search per leg	Done	Available trains loaded by origin/destination/date
Yacht search per leg	Partial	UI components exist; backend availability query for yacht legs unconfirmed
Results list per leg	Done	Options shown in a list per transport mode
Vehicle detail popup	Done	Detail view for each option before selecting
Real-time availability update	Missing	Results fetched once; no live refresh if a schedule fills up mid-session

3. Selection & Cart

Choosing a transport option for each leg and reviewing the full journey.

Sub-Feature	Status	Notes
Select a vehicle / schedule per leg	Done	Selection stored in <code>MultiModelBooking</code>
Cart review page (all legs + costs)	Done	<code>ReviewJourney</code> page shows all selected legs
Per-leg cost display	Done	Individual leg prices shown
Server-side total price validation	Missing	Total assembled client-side only — manipulation risk before payment
Seat selection — bus leg	Missing	Skipped in multi-model flow; no seat number assigned
Seat selection — train leg	Missing	Skipped; train seat conflict detection not applied
Remove / change a leg before payment	Partial	UI may allow re-selection; backend update confirmation unclear

4. Traveller Details

Collecting passenger information for the booking.

Sub-Feature	Status	Notes
Passenger name	Done	Captured in traveller details form
Passenger contact info	Done	Phone and email
Passenger ID number	Done	National ID / passport number
Multiple passengers per booking	Partial	Form appears single-passenger; group booking unclear
Luggage / baggage declaration	Missing	No cross-leg baggage summary or allowance display

5. Payment

Completing the financial transaction for the multi-leg booking.

Sub-Feature	Status	Notes
Payment page renders with booking summary	Done	Page loads and shows leg breakdown
Server-side total sent to PayHere	Partial	Unconfirmed; current total may come from client-side
PayHere payment initiation	Partial	Payment page exists; actual PayHere API call unconfirmed
Payment reference stored against booking	Partial	Must confirm <code>MultiModelBooking</code> stores PayHere transaction ID
Payment failure handling	Missing	No confirmed retry flow
Vendor approval workflow after payment	Done	Multi-leg booking routed to each provider for confirmation
Vendor payout split (multiple vendors)	Missing	No automated split; all revenue goes to one bucket

6. Booking Confirmation

What happens after payment is completed.

Sub-Feature	Status	Notes
Booking confirmation page — bus legs	Done	Confirmation page per transport type
Booking confirmation page — train legs	Done	Confirmation page per transport type

Booking confirmation page — yacht legs	Done	Confirmation page per transport type
Booking reference number	Done	Reference shown on confirmation
Unified itinerary PDF (all legs in one document)	Missing	No single-document itinerary; each leg needs its own ticket
Email confirmation to passenger	Partial	Confirmation page exists; email delivery unconfirmed
Vendor notification per leg	Missing	No fan-out notification to each provider after payment

7. Post-Booking Management

What passengers and vendors can do after confirmation.

Sub-Feature	Status	Notes
Booking history in client account	Partial	Multi-model bookings may not appear in the standard client dashboard
Cancel entire booking	Partial	No confirmed cancellation flow for multi-model bookings
Cancel an individual leg	Missing	No workflow to cancel one leg without affecting others
Modify a leg (change date / transport)	Missing	Not supported post-confirmation
Add a new leg after booking	Missing	Not supported
Refund handling on cancellation	Missing	No refund logic for multi-model bookings

8. Transport-Type Integration

How well each transport type is wired into the multi-model backend.

Sub-Feature	Status	Notes
Bus — schedule lookup	Done	Calls bus schedule query
Bus — booking creation	Partial	Booking record created; seat number not assigned
Bus — seat conflict check	Missing	Bus seat validation not applied in this flow
Train — schedule lookup	Done	Calls train schedule query
Train — booking creation	Partial	Booking record created; seat number not assigned
Train — seat conflict check	Missing	Train seat unique-constraint not applied in this flow

Yacht — availability lookup	Partial	UI exists; backend yacht availability controller unconfirmed
Yacht — booking creation	Partial	Unclear if controller logic for yacht leg booking is complete

Critical Pre-Launch Fixes

#	Fix	Impact
1	Server-side price validation — Calculate and validate the total on the server in <code>MultiModelBookingController</code> before initiating PayHere. Never trust a client-submitted total.	High — security / financial
2	PayHere integration — Confirm PayHere is called with the correct amount and that the transaction ID is stored against <code>MultiModelBooking</code> .	High — financial
3	Seat assignment for bus / train legs — Add seat selection steps or implement auto-assignment. Without seat numbers, operators cannot honour the booking.	High — operations
4	Yacht backend audit — Check <code>MultiModelBookingController</code> for yacht-leg booking logic. If missing, remove yacht from multi-model at launch.	High — reliability
5	Auth gate — Block unauthenticated users at the journey-plan step (not silently at checkout) to avoid losing planned journey data.	Medium — UX
6	Vendor fan-out notification — After payment, notify each vendor whose leg was booked.	Medium — operations